Mohamed "Oscar" Ghanem

Website | Oschart | maghanem | Publications
Saarbrücken, Germany | m.ghanem.mg16@gmail.com

First-Author Publications

- "The Shape of a Program: Path Signatures for Trace-to-Program Induction." ***NeurIPS 2026***.
- "Flowing Through States: Neural ODE Regularization for Reinforcement Learning." ***ICLR 2026***.
- "Learning Better Representations From Less Data For Propositional Satisfiability." ***NeurIPS 2024 (Spotlight)***.
- "FLoBC: A Decentralized Blockchain-Based Federated Learning Framework." **IEEE BCCA 2022**.

Employment

Doctoral Researcher February 2024 – Present
CISPA Helmholtz Center

- Develop neuro-symbolic techniques for formal reasoning domains.
- Designed NeuRes, a framework for learning to solve Boolean Satisfiability.
- Tutored the Neuro-Symbolic Computing 2024 seminar; reviewer for NeurIPS, ICLR & ICML (2025–2026).

Machine Learning Engineer October 2021 – December 2023
Spree3D

- Developed an automated evaluation suite for Virtual Try-On models with neural KPIs (e.g., structural similarity).
- Built an end-to-end deep learning pipeline for volumetric 3D hair generation from portrait images.
- Implemented and tweaked a state-of-the-art video segmentation model that served most of our machine learning pipelines.

Research Assistant June 2022 – April 2023
Max-Planck Institute: MPI-SWS

- Developed a PyTorch implementation for a meta-learning technique that outperformed most benchmarks.
- Built a unifying Python library for all prominent reward-shaping techniques following a **stable-baselines3** style.

Automated Reasoning Tutor November 2021 – February 2022
Max-Planck Institute: MPI-INF

- Conducted weekly tutorials on the foundations of automated theorem provers.

Software Engineer July 2019 – October 2020
Nabta Health Ltd.

- Built React Native features, native NFC sensor drivers (Android & iOS), and the REST APIs behind them.

Education

Technical University of Munich February 2024 – Present
Munich, Germany

- **PhD in Computer Science**, expected 2028. Advisor: Prof. Bernd Finkbeiner.

Saarland University October 2021 – January 2024
Saarbrücken, Germany

- **M.Sc. in Computer Science**, GPA: **1.3** (German scale, 1.0 = best)

The American University in Cairo September 2016 – June 2021
Cairo, Egypt

- **B.Sc. in Computer Engineering**, GPA: **3.97/4.0** (American scale); **Minor in Mathematics**, GPA: **4.0/4.0**
- Full-tuition PSSF scholarship (115k USD) for outstanding academic performance.

Languages and Technologies

- **Languages:** Python, Lean, C/C++, CUDA, Rust
- **Tools/Frameworks:** PyTorch, JAX, Docker